

An Infinite Essential-Singularity Ladder beyond the Local Hénon–Lind Counterterm

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Abstract

The odd reflection-packet Euler product of the full Hénon horseshoe and the Lind zeta of the full two-shift reverse action admit a unique local relative counterterm at their common entropy boundary. We determine what happens after that first cancellation. Writing $\Phi(x) = 2x/(1 - 2x^2)$, we prove the exact regrouping

$$\log \mathcal{Z}_{\text{orb}}(t, 1) = \sum_{m \geq 1} c_m \Phi(t^m), \quad c_m = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1 - p).$$

Every coefficient c_m is nonzero. The locally normalized relative germ therefore continues, on punctured branches, as

$$\log C_{\text{rel}}(t) = H_{\text{rel}}(1 - \sqrt{2}t) - \sum_{m \geq 2} c_m \Phi(t^m),$$

where H_{rel} is explicit and regular on the positive corridor. For every $m \geq 2$, the point $\rho_m = 2^{-1/(2m)}$ is an exponential essential singularity, and $\rho_m \nearrow 1$. Thus the first-boundary counterterm is locally exact but cannot globalize to a meromorphic finite-state or holomorphic trace-class Fredholm determinant on the unit disk. A punctured-domain infinite-rank model remains open; no arithmetic trace or Route-B claim is made.

1 The local bridge and the global question

For the full two-shift with reverse flip, Kim et al. [2, Example 4.3] give

$$\zeta_{\text{flip}}(t) = (1 - 2t^2)^{-1/2} \exp\left(\frac{2t + 3t^2}{1 - 2t^2}\right). \quad (1)$$

The conjugate full Hénon horseshoe [1] also carries the different odd marked-reflection product

$$\mathcal{Z}_{\text{orb}}(t, 1) = \prod_{\substack{n \geq 1 \\ n \text{ odd}}} (1 - t^n)^{-D_n}, \quad (2)$$

where D_n is the number of primitive marked reflection words of period n . This product is not the full infinite-dihedral Lind zeta.

Put $u = 1 - \sqrt{2}t$. The preceding local comparison proves that

$$C_{\text{rel}}(t) = u^{1/2} e^{-3/(4u)} \frac{\zeta_{\text{flip}}(t)}{\mathcal{Z}_{\text{orb}}(t, 1)} \quad (3)$$

extends holomorphically and nonvanishingly across $u = 0$ on a local branch. The power $1/2$ and exponential coefficient $3/4$ are unique. The present paper asks whether this local germ can be a meromorphic determinant on the whole unit disk.

2 A scalar-channel regrouping

Let

$$\Phi(x) = \frac{2x}{1 - 2x^2}. \quad (4)$$

All odd reflection words have generating function $\Phi(t)$, and primitive dilation Möbius inversion gives

$$E(t) := \sum_{\substack{n \geq 1 \\ n \text{ odd}}} D_n t^n = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \mu(k) \Phi(t^k). \quad (5)$$

Taking the logarithm of (2), first in its disk of absolute convergence, yields

$$\log \mathcal{Z}_{\text{orb}}(t, 1) = \sum_{r \geq 1} \frac{1}{r} E(t^r). \quad (6)$$

Theorem 2.1 (Exact channel decomposition). *For $|t| < 2^{-1/2}$,*

$$\log \mathcal{Z}_{\text{orb}}(t, 1) = \sum_{m \geq 1} c_m \Phi(t^m), \quad (7)$$

where

$$c_m = \frac{1}{m} \sum_{\substack{k|m \\ k \text{ odd}}} k \mu(k) = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1 - p). \quad (8)$$

In particular, $c_m \neq 0$ for every $m \geq 1$.

Proof. Insert (5) into (6) and group terms with $m = kr$. The coefficient of $\Phi(t^m)$ is

$$\sum_{\substack{k|m \\ k \text{ odd}}} \frac{\mu(k)}{m/k} = \frac{1}{m} \sum_{\substack{k|m \\ k \text{ odd}}} k \mu(k).$$

The sum is multiplicative on the odd radical of m , giving the Euler product in (8). Every factor $1 - p$ is nonzero. $\square$

The decomposition is more than a coefficient identity: each m is now an independent rational singular channel.

3 Continuation after the first cancellation

The $m = 1$ channel is precisely $\Phi(t) = 2t/(1 - 2t^2)$. Direct substitution into (1) gives the following exact regular remainder.

Proposition 3.1 (Relative continuation formula). *Set*

$$H_{\text{rel}}(u) = -\frac{1}{2} \log(2 - u) - \frac{3(2u - 3)}{4(u - 2)}. \quad (9)$$

On every compatible branch that begins at the origin and avoids the stated singular points,

$$\log C_{\text{rel}}(t) = H_{\text{rel}}(1 - \sqrt{2}t) - \sum_{m \geq 2} c_m \Phi(t^m). \quad (10)$$

The series is normally convergent on compact subsets of $|t| < 1$ that avoid the zeros of $1 - 2t^{2m}$, $m \geq 2$.

Proof. The logarithm of (3) is

$$\frac{1}{2} \log u - \frac{3}{4u} + \log \zeta_{\text{flip}}(t) - \log \mathcal{Z}_{\text{orb}}(t, 1).$$

Because $1 - 2t^2 = u(2 - u)$, subtracting the $m = 1$ channel gives exactly

$$\frac{1}{2} \log u + \log \zeta_{\text{flip}}(t) - \frac{3}{4u} - \Phi(t) = H_{\text{rel}}(u).$$

This proves (10) in the initial disk and hence by continuation. Also $|c_m| \leq 1$. On a compact set with $|t| \leq r < 1$, all sufficiently large terms satisfy $|\Phi(t^m)| \ll r^m$; the finitely many earlier denominators are bounded away from zero. This proves normal convergence off the punctures. $\square$

Notice that (9) is regular throughout the positive interval $0 < t < 1$. All later positive singularities therefore come from the packet channels, not from the source Lind factor.

4 The infinite essential-singularity ladder

For $m \geq 1$, define

$$\rho_m = 2^{-1/(2m)}, \quad v_m = 1 - t/\rho_m. \quad (11)$$

Then $1 - 2t^{2m} = 2mv_m + O(v_m^2)$ and $2t^m = \sqrt{2} + O(v_m)$.

Theorem 4.1 (Positive essential ladder). *For each $m \geq 2$, the continuation in (10) has*

$$\log C_{\text{rel}}(t) = -\frac{c_m}{\sqrt{2}m} \frac{1}{1 - t/\rho_m} + G_m(t), \quad (12)$$

where G_m is holomorphic near ρ_m . Consequently C_{rel} has an exponential essential singularity at every ρ_m , and

$$2^{-1/4} = \rho_2 < \rho_3 < \cdots < 1, \quad \rho_m \longrightarrow 1. \quad (13)$$

Proof. At the positive point ρ_m , the equation $1 - 2\rho_m^{2j} = 0$ holds only when $j = m$. Hence all channels except the m th are holomorphic there. The local expansion above gives

$$\Phi(t^m) = \frac{1}{\sqrt{2}m} \frac{1}{v_m} + O(1).$$

Insert this into (10). By (8), $c_m \neq 0$, so exponentiating the nonzero simple pole in (12) produces an essential singularity. Monotonicity and the limit in (13) are immediate. $\square$

The $m = 1$ singularity is exactly the one removed by the unique P71 counterterm. The theorem says that this cancellation is genuinely local: it reveals, rather than removes, all later channels.

5 Determinant obstruction and scope

Corollary 5.1 (No unit-disk meromorphic determinant). *The relative continuation C_{rel} is not meromorphic on the unit disk. In particular, it cannot equal a finite-dimensional determinant or a quotient of holomorphic trace-class Fredholm determinants on that disk.*

Proof. A meromorphic function has only poles as isolated nonremovable singularities. By theorem 4.1, C_{rel} has infinitely many isolated essential singularities inside the disk. Finite-dimensional determinants are holomorphic polynomials, and a quotient of holomorphic Fredholm determinants is meromorphic. $\square$

This is an object-specific obstruction, not a prohibition on every operator model. A transfer operator may live on a slit or punctured domain, use an infinite singular counterterm, or fail to be trace class at the ladder. No such model is constructed here. Nor do the channel indices m carry rational-prime or von-Mangoldt semantics. Route A gains an exact global analytic obstruction but no arithmetic advance; Route B remains unauthorized.

6 Executable certificate

The certificate verifies (8) in divisor-sum and Euler-product forms, compares (7) coefficientwise with the primitive Euler product through degree 48, and records the first 24 singular channels. An independent implementation reconstructs 64 channels without importing the main module. Eight tests pass in normal and optimized modes, six dependencies are hash-locked, and 25 claim mutations are rejected. The finite ledger certifies the algebra; theorems 2.1 and 4.1 prove the all- m statements.

References

- [1] Robert L. Devaney and Zbigniew Nitecki. Shift automorphisms in the Hénon mapping. *Communications in Mathematical Physics*, 67:137–146, 1979. doi: 10.1007/BF01221361.
- [2] Young-One Kim, Jungseob Lee, and Kyewon K. Park. A zeta function for flip systems. *Pacific Journal of Mathematics*, 209(2):289–301, 2003. doi: 10.2140/pjm.2003.209.289.